



OPEN Periapical lesion detection in periapical radiographs using the latest convolutional neural network ConvNeXt and its integrated models

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To overcome the limitation of a single classification model's inability to simultaneously identify multiple lesion targets within periapical radiographs, This study proposes YoCNET (Yolov5 + ConvNeXt), a novel deep learning integrated model. YoCNET leverages the target detection capability of Yolov5 and the image classification capability of ConvNeXt to achieve automatic segmentation of individual teeth and concurrent detection of periapical lesions across multiple teeth. A dataset of 1,305 periapical radiographs was used to train and validate the ConvNeXt and ResNet34 models, with an 8:2 split for training and validation. Deciduous teeth were excluded from the dataset. Furthermore, 717 individual teeth images were extracted from 200 previously unused periapical radiographs for integrated model validation. Evaluation metrics included accuracy, precision, sensitivity, F1 score, AUC (Area Under Curve), and a confusion matrix. The YoCNET integrated model demonstrated values of 90.93%, 98.88%, 85.30%, 0.9159, and 0.9757 for accuracy, precision, sensitivity, F1 score, and AUC, respectively. These metrics were superior to those achieved by the YoRNET (Yolov5 + ResNet34) integrated model, which recorded 80.47%, 83.78%, 82.16%, 0.8296, and 0.8822. The integrated model achieved high accuracy and efficiency in automatic teeth segmentation by Yolov5 and in automatically detecting multiple periapical lesions by ConvNeXt. YoCNET exhibited superior overall data performance, making it a more suitable deep learning integrated model for clinical applications.

Apical periodontitis is a prevalent inflammatory condition within dentistry, accounting for approximately 75% of radiolucent jaw lesions¹ and imposing a significant global health burden². Typically asymptomatic in its early stages, it is frequently identified during the radiographic follow-up of teeth post-root canal treatment³. This often results in delayed detection, missing the optimal treatment timeframe, which can lead to tooth loss and potential systemic implications for the patient⁴. Radiographically, apical periodontitis is characterized by a radiolucent area indicative of an expanded periodontal ligament space⁵, known as periapical lesions. Diagnostic approaches include Cone Beam Computed Tomography (CBCT), panoramic radiographs, and periapical radiographs. However, the high costs and radiation exposure associated with CBCT restrict its widespread use⁶. Periapical radiographs are considered the standard for identifying radiolucent periapical changes⁷, offering a cost-effective and detailed examination method that provides a comprehensive view of the affected tooth and adjacent tissues⁸. Compared to panoramic radiographs, periapical radiographs have a higher resolution, making them a common

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technique for diagnosing and treating periapical lesions⁹. Therefore, detecting periapical lesions on periapical radiographs is a routine task for dentists, but the determination of such lesions heavily relies on the dentist's knowledge and experience¹⁰, and is prone to fatigue during examination¹¹. These dependencies and drawbacks make automated and objective analysis methods an attractive option to assist young dentists who lack extensive knowledge and experience.

Demonstrated to offer accurate and swift disease diagnosis, deep learning enhances clinical decision-making and patient outcomes¹². Particularly within dentistry, CNNs have garnered increasing interest for their diagnostic capabilities^{13,14}. A notable study introduced a novel 3D CNN algorithm, PLA-Net, for the detection of periapical lesions in 4951 CBCT images¹⁵. The algorithm achieved an area under the receiver operating characteristic curve (AUC) of 0.98, aiding in improving the AUC values for diagnosing periapical lesions among dentists of varying experience levels, while also reducing the time taken for diagnosis. Additionally, research by Chen¹⁶ et al. employed a CNN model based on Faster R-CNNs for identifying and classifying dental caries, periapical periodontitis, and periodontitis in 2900 periapical radiographs. The average precision for detecting mild periapical lesions was below 0.2, moderate lesions had an average precision of 0.2–0.3, and severe lesions had 0.5–0.6. Setzer¹⁷ et al. reported a 93% accuracy in detecting periapical lesions in CBCT images using a deep learning approach. A recent study systematically assessed the effectiveness of deep learning models in detecting periapical lesions using 195 CBCT images of both the maxilla and mandible. The model demonstrated a sensitivity of 86.7% and a specificity of 84.3%. Notably, when very small lesions, identified by low periapical index scores, were excluded, the sensitivity improved to 90.4%¹⁸.

Current research on deep learning models for periapical lesion recognition predominantly relies on single models. These models suffer from two limitations. First, they necessitate manual segmentation of individual tooth regions in the image, increasing diagnostic time. Second, single classification models can only classify periapical lesions as positive or negative, hindering the simultaneous identification of multiple lesions in a single periapical radiograph. To address these limitations, this study propose a method utilizing an integrated model for automated segmentation of all tooth regions and concurrent detection of periapical lesions within periapical radiographs.

ConvNeXt, a novel deep learning framework for image classification, achieved an 87.8% Top-1 accuracy on the ImageNet dataset in 2022. To our knowledge, there have been no studies on ConvNeXt's application in identifying periapical lesions. This research intends to evaluate the performance of the image classification model ConvNeXt and the object detection model YOLOv5 integrated model YoCNET in detecting periapical lesions in periapical radiographs. Moreover, the ResNet series is currently a mainstream image classification algorithm, which has also achieved comparatively good results in classification tasks with its residual network architecture¹⁹. Therefore, this research selects the YoRNET integrated model as a comparison.

Materials and methods

Experimental design

The proposed approach for automated periapical lesion detection utilizes a multi-stage image processing pipeline (Fig. 1). The first stage involves image acquisition, where periapical radiographs are collected. Next, the collected radiographs are fed into the YOLOv5 object detection model. YOLOv5, a deep learning architecture, is employed to

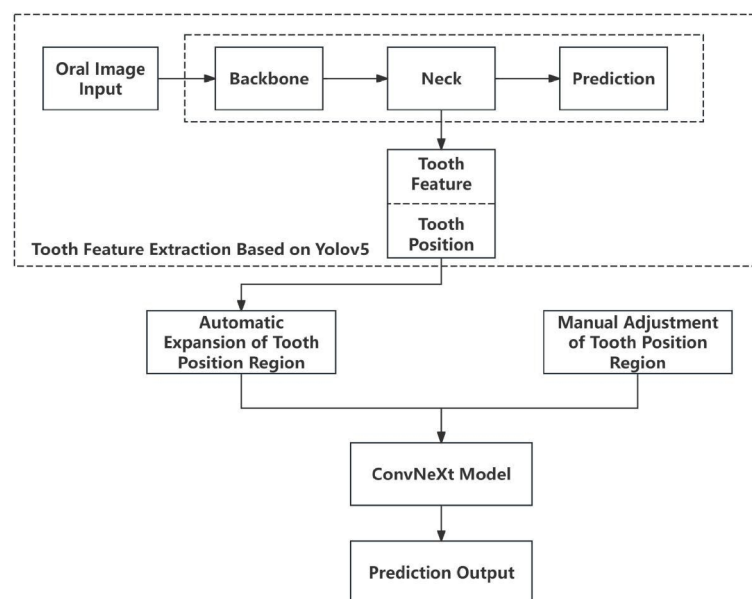


Fig. 1. Experimental flowchart overview: YOLOv5 segments images of individual teeth based on their features and positions, automatically expanding the selected tooth region. This selection can then be manually refined, before being passed to ConvNeXt for the final image classification task.

localize individual teeth within the images. The model accurately identifies each tooth and generates bounding boxes that encompass their corresponding regions. Following tooth detection, the selected region surrounding each identified tooth is automatically expanded. This step ensures that the target detection region encompasses not only the visible portion of the tooth but also the surrounding structures relevant for periapical lesion analysis. Finally, the expanded target detection regions are fed into the ConvNeXt deep learning model. ConvNeXt, a powerful CNN architecture, is tasked with classifying each region as either containing a periapical lesion or not (Fig. 2). The model outputs a classification probability for each region, indicating the likelihood of periapical lesion presence. In order to improve the interpretability of the model, this study utilized Gradient-weighted Class Activation Mapping (Grad-CAM)²⁰ to visualize the areas of periapical lesions.

Data acquisition and preprocessing

The datasets utilized for the training and validation of both individual and integrated models were derived from periapical radiographs acquired at Jinan Stomatological Hospital over the period from 2018 to 2023. To train YOLOv5 for individual tooth recognition, 600 root apex radiographs were used with a training image size of 640×640 pixels, 50 training epochs, and a batch size of 16. A total of 1,305 periapical radiographs were collected, comprising 650 positive and 655 negative cases. The dataset was split 8:2 for training and validation. A balanced dataset of 1,044 individual teeth periapical radiographs was used to train the ConvNeXt and ResNet models, consisting of 520 images with periapical lesions and 524 without. To improve model robustness, the images were augmented threefold through horizontal flipping, exposure adjustment, and sharpening. The final training set contained 1,560 positive and 1,572 negative images, totaling 3,132 images. The validation set remained unchanged, with 261 images (131 negative and 130 positive). To evaluate the performance of the integrated model, 200 previously unused periapical radiographs were selected as the validation set. From these, 717 individual teeth images were segmented, comprising 302 cases with positive periapical lesions and 415 cases with negative periapical lesions. All training images were resized to 640×320 pixels, with the number of epochs set to 50 and a batch size of 4. The training was conducted on a system running Ubuntu 22.04, equipped with dual RTX 3090 GPUs. All periapical radiographs used in this study were obtained with patients' informed consent.

Gold standard for diagnostic task

In establishing a gold standard for the diagnosis of periapical lesions, three oral radiologists with more than 15 years of experience each played a critical role. Utilizing the LabelImg image annotation tool, these experts annotated the periapical lesions within the images of individual teeth, specifically marking the regions of interest (ROIs) that exhibited periapical lesions. A consensus among at least two of the three doctors was requisite for validating the presence of periapical lesions in the images. The annotation files thus created were subsequently trained using PyTorch software. Written informed consent was obtained from all participating doctors. The study adhered to the ethical guidelines of the Declaration of Helsinki (1975 and 2013 revisions). Furthermore, all periapical radiographs were interpreted in a standardized radiology reading room.

Evaluation of diagnostic performance

Given that the target detection models in both integrated models were YOLOv5, differing only in the accompanying image classification models—ConvNeXt and ResNet34—this study adopted widely accepted evaluation metrics for image classification algorithms, such as accuracy, precision, sensitivity, F1 score, AUC, and the confusion matrix, to assess the performance of ConvNeXt, ResNet34, and their integrated models YoCNET and YoRNET. The definitions and formulas for each evaluation criterion (1–4) are detailed as follows:

Precision: The proportion of actual positives among all samples predicted to be positive for periapical lesions.

F1 Score: The harmonic mean of precision and recall, used to measure the balance between the model's accuracy and recall.

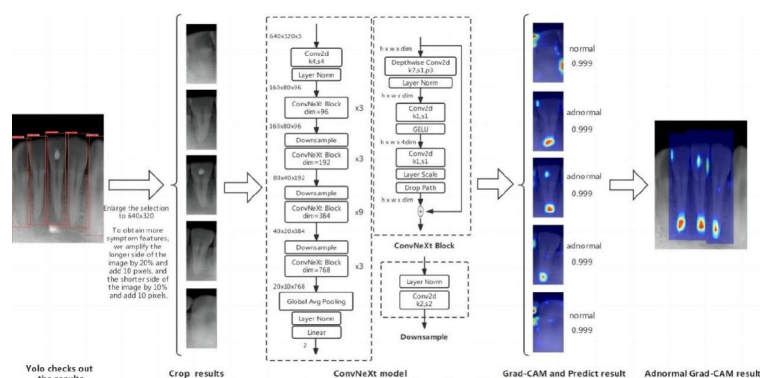


Fig. 2. The YOLOv5 model segments individual teeth and generates target detection areas. These areas are then input into the ConvNeXt model to detect periapical lesions in each tooth, and the detection results are generated simultaneously. The attention heat map displays the periapical lesion area, achieving the simultaneous detection of periapical lesions in all teeth in a single image.

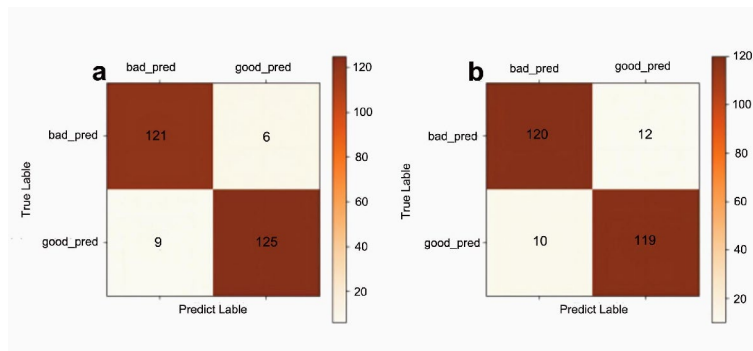


Fig. 3. Confusion matrices a and b are used to evaluate the performance of ConvNeXt (a) and ResNet34 (b), respectively. The top-left corner represents true positives (TP), the top-right corner false positives (FP), the bottom-left corner false negatives (FN), and the bottom-right corner true negatives (TN). ConvNeXt outperforms ResNet34 in accurately classifying both positive (TP) and negative (TN) cases of periapical lesions.

	Accuracy	Precision	Sensitivity	F1
ConvNeXt	94.25%	93.28%	95.41%	0.9433
ResNet	91.57%	92.24%	90.83%	0.9153

Table 1. Comparison of data between the single models ConvNeXt and ResNet34.

Accuracy: The proportion of correct predictions among all predictions.

Sensitivity (True Positive Rate): Identical to Recall, it represents the proportion of correctly predicted positive cases among all actual positive cases for periapical lesions.

AUC (Area Under the Curve): AUC measures the area under the ROC curve (Receiver Operating Characteristic curve), which is used to assess the performance of a classifier. The higher the AUC value, the better the performance of the model.

$$Precision = \frac{TP}{TP+FP} \quad (1)$$

$$F1 = \frac{Precision \times Recall}{Precision + Recall} \times 2 \quad (2)$$

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad (3)$$

$$Sensitivity = \frac{TP}{TP+FN} \quad (4)$$

In this study, TP represents true positives, which are the number of samples correctly predicted to have periapical lesions. FP stands for false positives, which are the number of samples incorrectly predicted to have periapical lesions. TN represents true negatives, which are the number of samples correctly predicted not to have periapical lesions. FN stands for false negatives, which are the number of samples incorrectly predicted not to have periapical lesions.

Results

The Confusion Matrix serves as a distinct tabular representation for assessing classification model performance, delineating the correlation between model forecasts and actual results. This matrix facilitates the computation of key metrics such as Accuracy, Precision, Sensitivity, and the F1 Score. The confusion matrices for ConvNeXt and ResNet34 models, evaluated on 261 periapical radiographs of individual teeth, are depicted in Fig. 3.

The ConvNeXt model demonstrated superior performance in detecting periapical lesions in individual teeth with an accuracy of 94.25%, precision of 93.28%, sensitivity of 95.41%, and an F1 score of 0.9433, surpassing the ResNet34's metrics of 91.57% in accuracy, 92.24% in precision, 90.83% in sensitivity, and an F1 score of 0.9153 (Table 1).

The performance metrics of ConvNeXt significantly outperform those of ResNet34, particularly in terms of sensitivity, where ConvNeXt demonstrates a marked advantage. This disparity, observed under uniform training strategies and parameter optimization, is likely attributable to the more efficient feature extraction and processing capabilities of ConvNeXt. Its sophisticated architecture enhances its sensitivity to subtle variations and strengthens its feature learning capacity. Consequently, these attributes enable ConvNeXt to more effectively identify teeth with lesions. The AUC offers a comprehensive evaluation of a model's performance across various classification thresholds, presenting a performance metric that is more robust than accuracy or other single-threshold indicators. AUC values span from 0 to 1, where higher values denote superior classification efficacy.

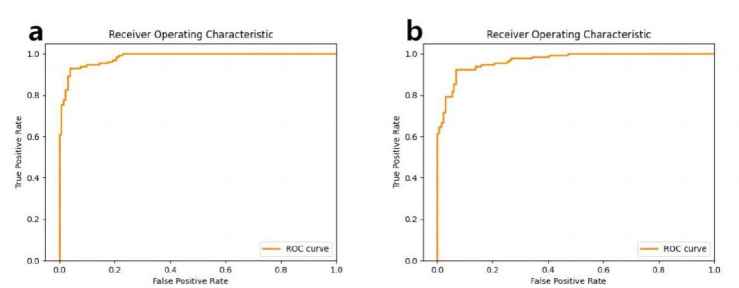


Fig. 4. The ROC curve serves as a graphical representation to evaluate the performance of binary classification models across diverse threshold settings. The x-axis denotes the False Positive Rate (FPR), defined in this study as the ratio of incorrectly classified negative periapical lesions as positive by the model. Conversely, the y-axis signifies the True Positive Rate (TPR), representing the ratio of periapical lesion positives accurately identified by the model. The ROC curve of (a) is closer to the top-left corner and maintains a high TPR, indicating that the model performs very well. It correctly classifies almost all positive samples while keeping the FPR very low. The ROC curve of (b) shows slight fluctuations during its rise, suggesting that the model may trade off between the FPR and the TPR at certain thresholds. Overall, the model in graph (a) performs better.

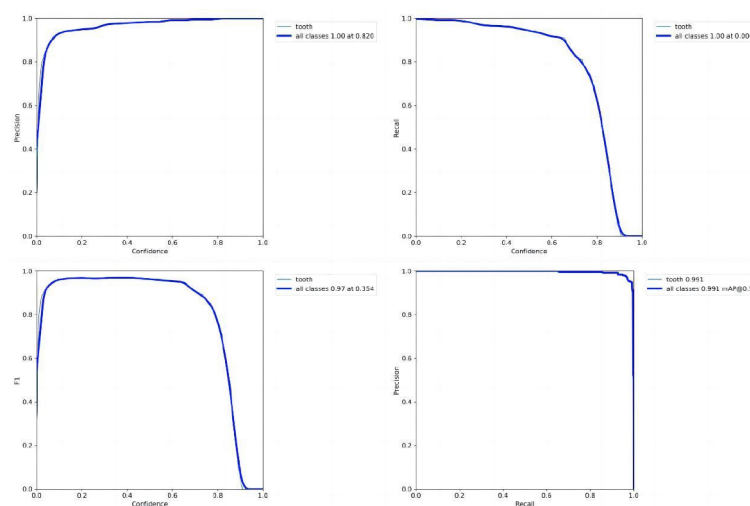


Fig. 5. The performance metrics of Yolov5 for identifying and segmenting individual teeth.

	Accuracy	Precision	Sensitivity	F1
YoCNET	90.93%	98.88%	85.30%	0.9159
YoRNET	80.47%	83.78%	82.16%	0.8296

Table 2. Across all performance metrics, YoRNET outperforms YoCNET, particularly in accuracy and precision.

The AUCs derived from the ROC curves for the ConvNeXt (Fig. 4a) and ResNet34 (Fig. 4b) models stand at 0.9820 and 0.9673, respectively.

Within the integrated model, Yolov5 achieves a mean Average Precision (mAP) of 0.991 (at an IoU threshold of 0.5) for the identification and segmentation of individual teeth in periapical radiographs (Fig. 5), demonstrating superior detection speed and real-time performance.

The integrated models YoCNET and YoRNET's periapical lesion detection outcomes on a consistent validation set, containing 717 periapical radiographs, are documented in Table 2. The corresponding confusion matrices are depicted in Fig. 6. The AUC values, derived from the ROC curves for the integrated models YoCNET and YoRNET, stand at 0.9757 and 0.8822, respectively (Fig. 7). With consistent training data, training regimen, and hyperparameters across models, YoCNET's superior performance relative to YoRNET across all evaluated metrics indicates that YoCNET is a more apt model architecture for the detection of periapical lesions.

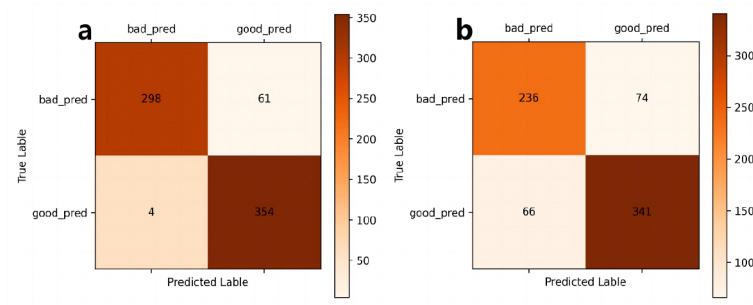


Fig. 6. Confusion matrices a and b are used to assess the performance of YoCNET (a) and YoRNET (b), respectively. YoCNET demonstrates significantly better accuracy than YoRNET in classifying both positive (TP) and negative (TN) cases of periapical lesions, with a particularly notable improvement in the classification of positive cases.

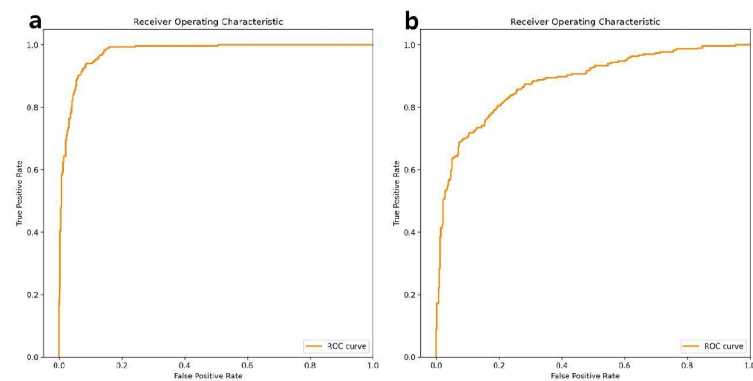


Fig. 7. The ROC curves for YoCNET (a) and YoRNET (b) demonstrate a positive correlation between TPR and FPR. Ideally, an ROC curve should be positioned as close to the top-left corner as possible, indicating the model's ability to maximize TPR while minimizing FPR in identifying periapical lesions.

The partial identification outcomes of the ensemble models are depicted in Fig. 8, illustrating original periapical radiographs images in column a, bounding results for all teeth by Yolov5 in column b, YoRNET integrated model results in column c, and YoCNET integrated model outcomes in column d. Both integrated models effectively detected periapical lesions. However, root canal fillings (row 1, column c; row 1, column d; row 2, column c; row 2, column d; row 4, column c; row 4, column d; row 5, column c) and high-density shadows in the dental crown ((row 4, column c) influenced the attention heatmaps to some extent. Overall, YoCNET exhibited superior performance, demonstrating a more focused attention heatmap and achieving better detection results for radiographs containing dental/root canal fillings and residual crowns/roots.

Discussion

Studies have demonstrated that among common dental diseases identified using deep learning, periapical lesions present the greatest challenge²¹. Clinically, early detection of periapical lesions followed by root canal treatment holds the promise of complete healing, or at the very least, a reduction in the size of the lesion²². Therefore, this research is highly significant. In this study, periapical radiographs taken at Jinan Stomatological Hospital over the past five years were randomly selected. It is noteworthy that even though diagnosing periapical lesions in teeth with root fillings presents challenges²³, periapical radiographs containing dental fillings, root canal fillings, or residual roots and crowns were not excluded in this study, as this more closely reflects real-world clinical scenarios. This study compared the performance of the latest deep learning model ConvNeXt with another mainstream model, ResNet34, in identifying periapical lesions in periapical radiographs, demonstrating the effectiveness and accuracy of the ConvNeXt model in detecting periapical lesions. The novelty of this study lies in the fact that the application of ConvNeXt to periapical lesion detection has never been reported before, filling this gap in the literature. Secondly, a single detection model that directly analyzes periapical images cannot effectively identify periapical lesions due to image overlap of adjacent structures and the presence of numerous low-density confusing structures around the apex, such as the maxillary sinus, mental foramen, and trabecular spaces. Segmenting images by individual teeth can reduce interference from neighboring tissues, and deep learning models can automatically complete the segmentation, improving efficiency. Therefore, this study achieved the combined use of multiple models, which overcomes the limitations of single model detection of

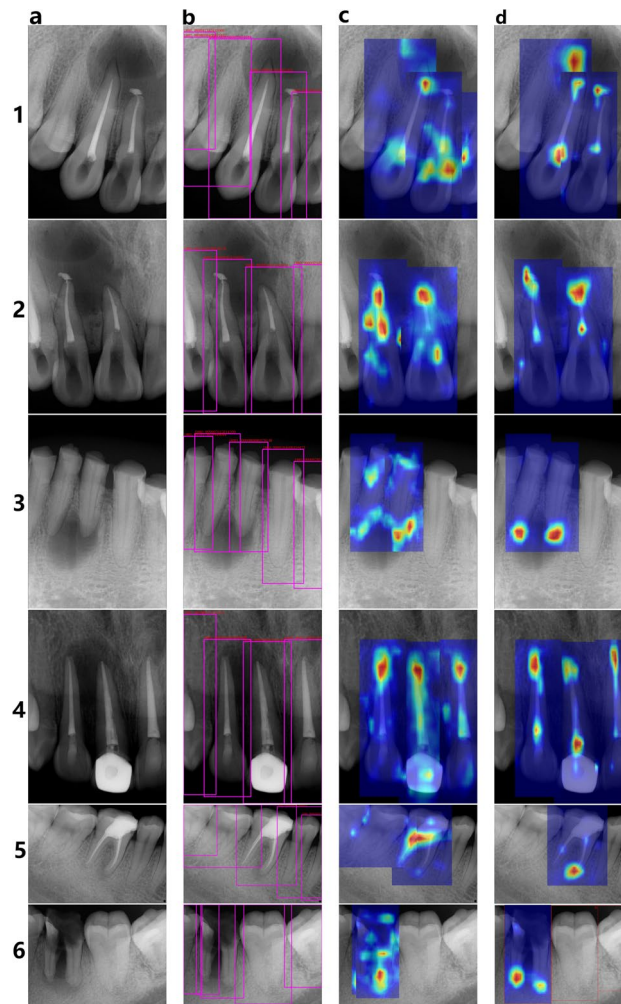


Fig. 8. Representative image comparisons of identification results between YoCNET and YoRNET. Column b images clearly demonstrate Yolov5's capability in identifying and segmenting individual tooth regions. In comparison to YoRNET, YoCNET demonstrates superior accuracy in detecting periapical lesions, characterized by more focused and distinct attention heatmaps. YoCNET's robustness to remnants, root canal fillings, and restorative crowns enhances its clinical applicability.

periapical lesions, achieving automatic segmentation of individual teeth and detection of periapical lesions in periapical images, thereby reducing human intervention.

The rapid development of artificial intelligence has improved the interpretation of medical imaging across various fields, including dentistry, especially through deep learning models and CNNs²⁴. A deep learning artificial intelligence model based on the U-Net framework segmented 63 periapical lesions in 47 panoramic X-rays, achieving a sensitivity, precision, and F1 score of 0.92, 0.84, and 0.88, respectively, at a 70% Intersection over Union (IoU)²⁵. Another study employing the U-Net for periapical lesion segmentation successfully identified 147 out of 180 periapical lesions in panoramic X-rays²⁶, with the model's F1 scores at IoU thresholds of 0.3, 0.4, and 0.5 being 0.828, 0.815, and 0.742, respectively. The U-Net-based convolutional neural network algorithm demonstrated high performance in the identification of periapical lesions. Both of the aforementioned studies were constrained by small datasets. In contrast, this study utilized a moderate number of periapical radiographs, applying minimal exclusion criteria (excluding only primary teeth). This approach reduces bias and enhances the model's robustness. The innovative advantage of this study lies in the combined use of the object detection model Yolov5 and the image classification model ConvNeXt. Yolov5, as a one-stage object detection algorithm, offers rapid detection, making it well-suited for efficiently screening large volumes of periapical radiographs. Moreover, it exhibits strong robustness to scale variations and occlusions, enabling reliable detection of periapical lesions even in complex backgrounds. ConvNeXt is a new deep learning model for image classification that based on the ResNet50 model, and improved in the manner of the Swin Transformer framework²⁷. It combines the traditional advantages of CNNs with the latest developments in Transformer models for visual tasks, achieving higher efficiency and accuracy. It outperforms the Swin Transformer in terms of accuracy and scalability. ConvNeXt can fully utilize convolutional kernels of different sizes to extract multi-scale features and has fewer parameters. This gives ConvNeXt higher accuracy and better generalization capability in image classification tasks. Yolov5 is well-

suited for detecting lesions of varying sizes, while ConvNeXt excels in extracting both local and global features from periapical radiographs, facilitating the detection of small or obscure lesions. The combined use of these models takes advantage of their complementary strengths, effectively detecting periapical lesions of various sizes, shapes, and locations. In clinical diagnosis, this combined approach enables the detection of periapical lesions in all teeth within a single periapical radiograph simultaneously, without requiring the diagnostician to manually segment individual teeth. This ensures diagnostic efficiency and accuracy, demonstrating significant potential for application in medical image analysis. One limitation of this study is that the opinions of three experts were used as the diagnostic standard for model training. Although the diagnosis of periapical lesions required the agreement of at least two experts, this could still introduce a degree of subjectivity. The knowledge, experience, and diagnostic criteria of different experts can significantly influence their judgments, often resulting in inconsistent opinions. Future research should focus on developing clear, standardized diagnostic protocols to ensure consistency in the annotation of periapical lesions, ultimately facilitating expert consensus. Additionally, the study only collected periapical radiographs from Jinan Stomatological Hospital and did not conduct a multicenter study. Medical institutions are the source of model data, but due to patient privacy concerns, data is not shared between institutions. Therefore, it is often only possible to collect data from a single institution, leading to a smaller dataset for model training and potential biases in the data distribution, with patient data but lacking data from healthy individuals. These limitations may cause the model to become overly reliant on the specific features of the single center, making it difficult to generalize to cases from other regions or hospitals. In a single-center dataset, annotations are usually provided by the same group of experts, which can result in consistency issues. Consequently, models trained on such data may struggle to accommodate broader variations in annotation practices. Future research will focus on optimizing deep learning algorithms and training strategies for image recognition by collecting periapical radiographs from multiple institutions using different X-ray machines. This approach will prevent the model from becoming overly dependent on the specific characteristics of a single center's data, allowing it to learn more diverse features. As a result, the model will be able to maintain high accuracy when encountering new, unseen data. In practical applications, the model will be adaptable to various devices and imaging conditions. Developing a deep learning model that combines automated individual tooth image segmentation with accurate diagnosis of periapical lesions, whether as a joint model or an integrated model, will be the focus of our future research. In clinical diagnosis, this combined approach improves the detection efficiency of early, subtle changes in periapical lesions. Accurate early diagnosis of periapical lesions can reduce the likelihood of the damage spreading to adjacent tissues or the root canal system, thus avoiding more invasive treatments, such as tooth extraction. This not only lowers treatment costs but also helps preserve the patient's dental health. By strengthening collaborations among different hospitals and institutions and collecting data from various sources, external validation can be conducted using independent datasets, enabling a more objective evaluation of the model's performance and providing feedback for continuous model optimization. Based on the validation results, researchers can refine the model, thereby continuously improving its performance and applicability. Successful external validation will enhance the credibility of the model for clinical applications. Thus, the application of artificial intelligence and deep learning in dentistry has a long way to go. Most importantly, the advancement of artificial intelligence is meant to better assist human doctors in diagnosis and decision-making, not to replace them.

Conclusion

The advanced deep learning model ConvNeXt demonstrates exceptional performance in identifying periapical lesions in periapical radiographs. This study proposed deep learning integrated model, YoCNET, enables automated segmentation of individual teeth in periapical radiographs and simultaneously achieve high-accuracy detection of periapical lesions in all teeth in a single periapical radiograph.

Data availability

The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

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Author contributions

J.L., X.H.L. and Y.S. wrote the main manuscript text and Y.Z.G., K.X.P. and C.R.J. prepared Figs. 1–8. H.H.J., Y.D. and X.J.Y. were responsible for the design and supervision of this study, and they also performed the final review and revision of the manuscript. All authors reviewed the manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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